SYSTEMS ARCHITECTURE & GREEN MULTI-AGENT PROTOCOLS · CANONICAL RELEASE V5.4.4

TOAI-elegant-lang: A Tri-Script Orthogonal RISC Wire Protocol, Adversarial Corpus Audit, and Empirical Comprehension Harness for Multi-Agent Communication

Koji Toriyama (PhenoX) · Principal Systems Architect, PhenoX AI Alliance, Kawasaki, Japan

Repository & Reproducibility Artifact: https://github.com/PhenoX-AI-Alliance/TOAI-elegant-lang | **PyPI:** `pip install toai-elegant-lang==5.4.4`

ABSTRACT

Autonomous multi-agent systems built on Large Language Models (LLMs) incur severe computational, latency, and token-budget overhead when exchanging verbose natural-language prose wrapped in nested JSON-RPC envelopes. Compressing agent-to-agent (A2A) messages exposes a fundamental systems trilemma among *wire token efficiency*, *syntactic, numeric, and causal execution fidelity*, and *human auditability*. This paper presents **TOAI-elegant-lang (v5.4.4)**, a zero-dependency, deterministic compiler and wire protocol inspired by the Rosetta Stone that aligns multi-agent state transitions across three synchronized representations: (1) **Demotic Wire Script**, a compact causal wire syntax (`@CTX:... :: §S01 ... §X64`); (2) **Sacred Expanded Script**, a deterministic macro- and primitive-expanded canonical form (`§GAIA_WARMTH_HARMONY`); and (3) the **Rosetta Audit Layer**, which attaches natural-language definitions and domain invariants for human inspection. Under `o200k_base` and `cl100k_base` Byte-Pair Encoding (BPE) tokenizers, each canonical §-prefixed identifier strictly occupies 3 BPE tokens (`'§' + Prefix + Digits`), while the 256-primitive + 256-macro codebook fits inside a $\sim$1,000-token system prompt header. Evaluated across standard A2A JSON serializers (`benchmarks/fair_bench.py`), Demotic Wire averages **73.2 BPE tokens/message**, achieving a **79.1% token reduction vs. Verbose A2A JSON** (349.8 tokens), **55.7% vs. Structured Tool JSON** (165.2 tokens), and **41.0% vs. Minified Key JSON** (124.2 tokens) while maintaining **100.0% downstream AST/task execution accuracy** (outperforming `LLMLingua`-style perplexity token pruning at $\rho = 0.35$, where numeric/task recovery degrades to 61.6%) and breaking even within **4–20 messages uncached** or **1–2 messages under 90% prompt caching**. We introduce a non-circular **Independent Corpus Alignment Auditor** (`src/audit_corpus_alignment.py`) validated against a human gold-standard sample ($n = 60$, $p_o = 93.33\%$, **Cohen's** $\kappa = 0.8611$). To eliminate circular validation, receiver comprehension is evaluated on a **strictly disjoint** $N = 112$ **Holdout Benchmark** (`corpus/holdout_comprehension.jsonl`, verified at **0.0% wire/scenario overlap**): Sacred expansion raises 4-choice blind accuracy from **44.6% (50/112)** on Compact Wire (Cond A) to **84.8% (95/112)** on Sacred Names (Cond B, exact McNemar $p = 1.36 \times 10^{-9}$, OR = 7.43, Cohen's $h = +0.878$) and **93.8% (105/112)** with Rosetta Glossary (Cond C, $p = 6.35 \times 10^{-3}$ vs. B), while the length-matched Shuffle Control negative baseline collapses to **24.1% (27/112)** (chance = 25.0%, $p = 3.59 \times 10^{-17}$ vs. B). Finally, we formalize origin attestation directly on Page 3 with a game-based EUF-CMA reduction bounding k-turn internal-counter HMAC-SHA256 forgery by $10^{-4k} + \epsilon_{\text{PRF}} + \epsilon_{\text{coll}}$.

1. Introduction and Multi-Agent Communication Trilemma

As Large Language Model (LLM) applications evolve from single-turn assistants into multi-agent workflows—e.g., Anthropic Model Context Protocol (MCP) [1], Google Agent-to-Agent (A2A) JSON-RPC 2.0 orchestrators [2], and AutoGen [5]—inter-agent communication overhead has become the dominant bottleneck in latency, context saturation, and KV-cache footprint. In conventional topologies, agents coordinate state transitions by generating conversational prose inside nested JSON schemas. While human-readable, schema keys and punctuation delimiters waste token budgets and scale attention FLOPs quadratically. Compressing inter-agent messages exposes a strict **Multi-Agent Communication Trilemma**:

1. **Wire Token Efficiency:** Minimizing BPE subword tokens per transition and amortizing shared schemas across sessions.
2. **Syntactic, Numeric, and Causal Execution Fidelity:** Preserving structured parameters (e.g., leading-zero strings `"007"`, embedded commas `"a, b"`, floats `150.50`), transition probabilities $p \in [0, 1]$, costs $c \in [0, 0.9999]$, and fault edges (`!->`) for downstream AST execution.
3. **Human Auditability and Cross-Model Portability:** Allowing humans and heterogeneous LLM families (GPT, Claude, Gemini, Llama) to inspect and exchange packets over standard text APIs without opaque binary blobs or model-specific latent adapters.

Existing approaches sacrifice at least one vertex (Table 1). Perplexity-based prompt compression (LLMLingua [4]) drops tokens based on LM perplexity, which breaks JSON syntax delimiters and drops numeric thresholds at high compression ratios ($\rho \leq 0.35$, Section 5). Conversely, emergent MARL discrete codes [6] or continuous latent vectors [7] sacrifice human auditability and fail across commercial text-only APIs. State-delta diffs reduce payload size on homogeneous repeated states but introduce desynchronization risks under network retries and still pay JSON overhead on multi-domain handoffs; crucially, TOAI-elegant-lang is stateless on the wire and orthogonal to state-delta encoding (deltas can be nested directly within Demotic `[k:v]` payloads). Flat key-value templates (`KEY=VAL`) reduce raw tokens but destroy multi-branch causal execution graphs and fault-recovery semantics.

To address this trilemma, we present **TOAI-elegant-lang (v5.4.4)**, a zero-dependency Python compiler and wire protocol built on an orthogonal Reduced Instruction Set Computer (RISC) architecture and Tri-Script Rosetta design. Crucially, release `v5.4.4` incorporates multi-AI adversarial audits (ChatGPT and Claude) and external peer review, replacing provisional estimates with a strictly disjoint $N = 112$ holdout benchmark, complete downstream execution accuracy comparisons, and a formal game-based cryptographic reduction.

Table 1: Architectural Comparison of Multi-Agent Communication Paradigms (Verified Standards & Repository Baselines)

Communication Paradigm	Wire Encoding	Causal AST & Guards	Deterministic Round-Trip	Cross-API Portability	Human Auditability & Operational Characteristics
Verbose A2A / MCP JSON-RPC [1, 2, 5]	Pretty JSON + NL Prose	Yes (Verbose)	100.0%	100% (Text API)	High readability; highest token cost (349.8 tok/msg; baseline).
Structured Tool JSON (`fair_bench.py`)	Standard JSON Schema	Yes (JSON Objects)	100.0%	100% (Text API)	Lossless schema; repeated JSON keys/syntax (165.2 tok/msg).
Minified Key JSON (`fair_bench.py`)	Short-Key JSON (`"c","s"`)	Yes (Abbrev. JSON)	100.0%	100% (Text API)	Abbreviated keys require schema lookup (124.2 tok/msg).
Simplified Flat Template (`KEY=VAL`)	Delimited Flat String	**No (Lossy)**	Lossy (0% AST)	100% (Text API)	Lowest raw tokens (34.5 tok/msg), but loses causal edges & 75% params.
Binary Protobuf [8] (Base64 Text Wire)	Base64-Armored Octets	Schema-Defined	100.0% (External)	Requires Tool/Decode	Opaque Base64 string in prompt context; BPE subword fragmentation.
Perplexity Token Pruning (LLMLingua [4])	Pruned Natural Text	Unstructured	Lossy (0% JSON)	100% (Text API)	Designed for prose/RAG; breaks JSON syntax; 61.6% task acc at $\rho = 0.35$.
Emergent MARL Codes [6, 7]	Learned IDs / Vectors	Implicit in Policy	Model-Dependent	Homogeneous Only	Opaque latent symbols; not directly interpretable by humans.
TOAI-elegant-lang (v5.4.4 Ours)	**Demotic RISC Wire (`§ID`)**	**Yes (Native LL(1) AST)**	**100.0% Lossless**	**100% (Text API)**	**Tri-Script (Demotic / Sacred / Rosetta); 73.2 tok/msg; 100% Task Acc.**

2. System Architecture, LL(1) Grammar, and Quote-Aware Stateful Codec

2.1 Tri-Script Architecture and 8-Bit Orthogonal RISC Lexicon Hierarchy

TOAI-elegant-lang decouples machine transmission from human inspection via three synchronized layers (Table 2):

1. **Demotic Wire Script (Compact Machine Layer):** Transmitted between agents. Combines context header (`@CTX:DOMAIN`), 1-byte logical core opcodes (`§S01`–`§H32` across 8 categories: System, Data, Governance, Fault, Execution, Audit, Gaia, Logic; exactly 256 primitives in `core_lexicon.json`), 256 active higher-order macros (`§X01`–`§X969` in `macro_lexicon.json`), typed payloads (`[k:v]`), and causal operators (`-[p:..., c:...]->`, `=>`, `!->`).
2. **Sacred Expanded Script (Canonical Symbolic Layer):** Deterministic expansion via `codec.decode(wire)` expanding opcodes to canonical uppercase snake-case identifiers matching strict `^§[A-Z][A-Z0-9_]*$` (e.g., `§GAIA_WARMTH_HARMONY`).
3. **Rosetta Audit Layer (Human Governance Layer):** Maps every symbol to concise natural-language definitions for compliance auditing (`rosetta_layer.json`) and Condition C glossary injection.

Exact BPE Token Accounting: Under `o200k_base` and `cl100k_base`, the section sign `'§'` (`U+00A7`) is tokenized separately from following ASCII characters. In all reported benchmarks, every canonical §-prefixed opcode (`§S01`, `§G48`, `§X64`) is strictly counted as **3 BPE tokens** (`'§'` + `Prefix` + `Digits`), confirming that our 73.2 tok/msg average includes full token penalties.

Table 2: Tri-Script Representation and Lexicon Hierarchy in TOAI-elegant-lang v5.4.4

Architectural Layer	Source / Capacity	Canonical Pattern / Example	Validation & Runtime Role in v5.4.4
1. Demotic Wire Script (Core Primitives)	`core_lexicon.json` (256 Primitives)	`§S01..§L32`, `§G48`, `§C20` (3 BPE tok/symbol)	Immutable 256-primitive base vocabulary; validated at startup and by `test_17` for bijective mapping.
1b. Demotic Wire Script (Active LRU Macros)	`macro_lexicon.json` (256 Active + Archive)	`§X01..§X969` (e.g., `§X64`, `§X354`; 3 BPE tok/symbol)	Depth-1 macro expansion into core primitives; pinned core macros prevent LRU thrashing.
2. Sacred Expanded Script (Canonical AST)	`codec.decode()` Output	`^§[A-Z][A-Z0-9_]*$` (e.g., `§GAIA_WARMTH_HARMONY`)	Zero lowercase or CJK corruption permitted; strictly enforced by unit test `test_17` across all files.
3. Rosetta Audit Layer (Human Definitions)	`rosetta_layer.json` & `definitions`	Natural-language semantic specification per symbol	Used for human governance inspection, `audit_corpus_alignment.py`, and `eval_comprehension.py --cond C`.
Inline Structured Payload (`[k:v]`)	Quote-Aware Scanner (`_parse_payload_block`)	`[pair:"USD/JPY", stop:150.5, code:"007", note:"a, b"]`	Stateful character scanner preserving quoted strings, leading zeros, commas, brackets, and escapes.

2.2 Conflict-Free LL(1) EBNF Grammar and Quote-Aware Stateful Payload Parser

To guarantee deterministic $\mathcal{O}(L)$ parsing without regex backtracking bugs or payload corruption, `v5.4.4` implements a conflict-free LL(1) grammar paired with a stateful character-level quote/escape scanner in `src/codec.py`:

```
packet = [ "@CTX:", identifier, "::" ], expr_node, { ( causal_arrow | "=>" | "!->" | ">" | "?" ), expr_node } ;
expr_node = ( "(", packet, ")" | "{", packet, "}" | "§", ( risc_id | macro_id | sacred_name ) ), [ "[", kv_pair, { ",", kv_pair }, "]" ] ;
kv_pair = identifier, ":", ( '"', { ( utf8_char - ( '"' | '\' ) ) | '\"' | '\\' }, '"' | float_lit | int_lit | bool_null_id ) ;
causal_arrow = "-[", [ "p:", float_val, "," ], ( "c:" | "cost:" ), cost_mantissa, "]->" ; (* cost_mantissa: exact 4 digits *)
```

The parser strictly distinguishes quoted strings (`"007"` → `str`) from unquoted numbers (`007` → `int`), isolates delimiters inside quotes (`note:"a, b"`, `expr:"x]y"`), and enforces explicit OOV policies (`on_oov="error" | "keep" | "nearest"` with an audit log in `EncodedWireResult.substitutions`, verified by tests `test_07`–`test_15`).

3. Formal Threat Model and Game-Based EUF-CMA Proof of Origin Attestation

3.1 Unobservable Internal-Counter Attestation Construction (Zero Timestamp Jitter)

To prevent an active Dolev-Yao adversary $\mathcal{A}$ from injecting forged commands or observing rejection-sampling loops under an adaptive Chosen-Message Attack (EUF-CMA), TOAI-elegant-lang embeds an origin tag directly into the 4-digit causal cost mantissa $c_n \in [0.0000, 0.9999]$. Let $S_0 \in \{0,1\}^{256}$ be the ephemeral session key established via X25519 ECDH and HKDF-SHA256, $H_0 = \text{SHA256}(\texttt{"TOAI_TELEPATH_GENESIS_V541"} \,\|\, \text{SHA256}(\texttt{core_lexicon.json}) \,\|\, \text{SHA256}(\texttt{macro_lexicon.json}))$, and $H_{n-1} = \text{SHA256}(H_{n-2} \,\|\, \texttt{0x00} \,\|\, \text{wire}_{n-1})$. For monotonic timestamp $t_n > t_{n-1}$, sender and verifier compute the 16-bit value $Y_n \in \{0, \dots, 59{,}999\}$ for public confidence mantissa $P_n = \lfloor p_n \times 10^4 \rfloor \in \{0, \dots, 9999\}$ via an **internal, non-transmitted rejection counter** $r_n \in \{0, 1, \dots\}$:

$$r_n = \min\Big\{ r \in \mathbb{N}_0 : \big(P_n \oplus \text{trunc}_{16}(\text{HMAC-SHA256}(S_0,\, H_{n-1} \,\|\, \texttt{0x00} \,\|\, \text{dom}(t_n, r)))\big) < 60{,}000 \Big\} \tag{1}$$

where $\text{dom}(t_n, 0) = t_n$ and $\text{dom}(t_n, r) = t_n \,\|\, \text{uint8}(r)$ for $r \geq 1$. Verifiers evaluate:

$$T_n = \text{trunc}_{16}\big(\text{HMAC-SHA256}(S_0,\, H_{n-1} \,\|\, \texttt{0x00} \,\|\, \text{dom}(t_n, r_n))\big), \qquad \big((P_n \oplus T_n) \bmod 10{,}000\big) = C_n \tag{2}$$

Because bitwise XOR is a bijection on $\{0, \dots, 65{,}535\}$, each candidate $P_n \oplus T_n^{(r)}$ is uniform with single-draw acceptance $60{,}000/65{,}536 = 91.55\%$ ($\mathbb{E}[r_n + 1] = 1.0923$). For $r_1 = 0$ (Table 3), $Y_1 = 9841 \oplus 9247 = 622 < 60{,}000$, yielding $C_1 = 622 \bmod 10{,}000 = 622 \implies c_1 = 0.0622$.

3.2 Formal Security Reduction ($\text{Game}_0 \to \text{Game}_2$) and Congruence Cardinality Proof

Theorem 1 (Exact EUF-CMA Forgery Bound Subsuming Direct Tag Guessing and Modular Aliases). *Let $\mathcal{A}(\tau, Q, k)$ be an adaptive EUF-CMA adversary making at most Q signing queries to MAC oracle $\mathcal{O}_{MAC}(S_0, \cdot)$ and outputting a k-turn unqueried forgery trajectory $\{(wire^*_{n+j}, t^*_{n+j}, P^*_{n+j}, C^*_{n+j})\}_{j=1}^{k}$ with $t^*_{n+j} > t^*_{n+j-1}$. Under the $(\tau', Q_{PRF}, \epsilon_{PRF})$-PRF security of HMAC-SHA256 (where $\mathbb{E}[Q_{PRF}] \leq 1.0923(Q + k)$) and (τ', ϵ_{coll})-collision resistance of SHA-256:*

$$\text{Adv}^{\text{EUF-CMA}}_{\text{TOAI}}(\mathcal{A}) \leq \left(\frac{6}{60{,}000}\right)^k + \epsilon_{\text{PRF}} + \epsilon_{\text{coll}} = 10^{-4k} + \epsilon_{\text{PRF}} + \epsilon_{\text{coll}} \qquad \left(P^{\text{unrej}}_{\text{forge}}(k) \leq \left(\frac{7}{65{,}536}\right)^k + \epsilon_{\text{PRF}} + \epsilon_{\text{coll}}\right) \tag{3}$$

Proof. We define three security games $\text{Game}_0 \to \text{Game}_1 \to \text{Game}_2$, letting W_i denote $\mathcal{A}$'s winning event in Game_i:

- **Game_0 (Real EUF-CMA Experiment):** Challenger samples $S_0 \xleftarrow{\$} \{0,1\}^{256}$, answers Q queries via Eqs. (1)–(2), and evaluates $\mathcal{A}$'s k-turn forgery. By definition, $\Pr[W_0] = \text{Adv}^{\text{EUF-CMA}}_{\text{TOAI}}(\mathcal{A})$.
- **Game_1 (Transcript Collision Abort):** Challenger aborts if two distinct wire histories ever produce matching chain digest $H = H'$. Any abort yields a SHA-256 collision within $\tau' \approx \tau + \mathcal{O}(Q + k)$, hence $|\Pr[W_0] - \Pr[W_1]| \leq \epsilon_{\text{coll}}$.
- **Game_2 (PRF Replacement):** We replace $\text{HMAC-SHA256}(S_0, \cdot)$ with a lazy uniform random function $\mathcal{R} : \{0,1\}^* \to \{0,1\}^{256}$. By PRF reduction over at most Q_{PRF} evaluations, $|\Pr[W_1] - \Pr[W_2]| \leq \epsilon_{\text{PRF}}$. Because digests are collision-free (Game_1) and timestamps strictly monotonic, every input $X^{(r)}_{n+j} = H^*_{n+j-1} \,\|\, \texttt{0x00} \,\|\, \text{dom}(t^*_{n+j}, r)$ during forgery verification is strictly fresh (never previously queried).

- **Residue Class Analysis in Game$_2$:** For each forged turn $j \in \{1,\dots,k\}$: (1) If $C^*_{n+j} \notin \{0,\dots,9999\}$, the LL(1) parser rejects deterministically ($\Pr = 0$). (2) If $C^*_{n+j} \in \{0,\dots,9999\}$, because $T \mapsto P^*_{n+j} \oplus T$ is a bijection and $\mathrm{trunc}_{16}(\mathcal{R}(X^{(r)}_{n+j}))$ is uniform, the accepted draw $Y_{n+j} = P^*_{n+j} \oplus T^{(r_{n+j})}_{n+j}$ is uniform on $\{0,\dots,59{,}999\}$. The winning congruence set has exact cardinality $|\mathcal{W}(C^*_{n+j})| = 6$:

$$\mathcal{W}(C^*_{n+j}) = \{C^*_{n+j} + 10{,}000m : m \in \{0,\dots,5\}\} \subset \{0,\dots,59{,}999\} \implies \Pr[Y_{n+j} \in \mathcal{W}(C^*_{n+j})] = \frac{6}{60{,}000} = 10^{-4} \tag{4}$$

subsuming Direct Tag Guessing ($m = 0$, prob 1/60,000) and Modular Aliases ($m \in \{1,\dots,5\}$, prob 5/60,000). Across k independent turns, $\Pr[W_2] = (6/60{,}000)^k = 10^{-4k}$. Without rejection sampling ($Y_{n+j} \in \{0,\dots,65{,}535\}$), residues $C^* \le 5{,}535$ admit a 7th alias ($m = 6$), bounding forgery by $\max|\mathcal{W}_{\text{unrej}}|/65{,}536 = 7/65{,}536 \approx 1.068 \times 10^{-4}$. Non-transmitted r_n adds ≈ 0.014 μs CPU variance, leaking negligible timing side-channel signal ($\ll 10^4$ μs network/LLM jitter). □

Table 3: Reproducible Cryptographic Test Vector for Equation (2) (Verifiable via Python `hashlib`/`hmac`, $r_1 = 0$)

Parameter / Verification Step	Canonical Hexadecimal / Decimal Value (Bitwise XOR Precedes mod 10,000)
Secret Seed S_0 (256-bit hex)	`7f83b1657ff1fc53b92dc18148a1d65dfc2d4b1fa3d677284addd200126d9069`
Session Anchor H_0 (SHA256)	`f3c515b25afe4f7ed4ca735b85fbfe50b083c7af4b8ec8c107c0e99438f02998` (`SHA256(b"TOAI_TELEPATH_GENESIS_V541")`)
Turn-1 Timestamp t_1 (UTF-8, $r_1 = 0$)	`2026-11-06T12:00:00Z` (HMAC-SHA256(S_0, $H_0 \,\|\|$ `b"\x002026-11-06T12:00:00Z"`) $\to T_1 =$ `0x241f` = **9247**)
Step 1: 16-Bit Bitwise XOR	Given $p_1 = 0.9841 \implies P_1 = 9841$: $Y_1 = (P_1 \oplus T_1) = (9841 \oplus 9247) =$ **622** $< 60{,}000$ ($r_1 = 0$ accepted)
Step 2: Modulo 10,000 Reduction	$C_1 = Y_1 \bmod 10{,}000 = 622 \bmod 10{,}000 =$ **622** $\implies c_1 = C_1/10^4 =$ **0.0622**

4. Multi-AI Adversarial Audit, Non-Circular Corpus Alignment, and Holdout Disjointness

4.1 Full Disclosure of Multi-AI Adversarial Audit Findings (v5.4.1 → v5.4.4)

Table 4 discloses every defect identified during blind audits across OpenAI ChatGPT and Anthropic Claude execution sandboxes through release `v5.4.4`, all permanently verified by automated test suites.

Table 4: Summary of Multi-AI Adversarial Audit Findings and Resolutions in Canonical Release v5.4.4

Audit Defect ID	Identified Vulnerability / Discrepancy	Deterministic Engineering Resolution in `v5.4.4` & Unit Test
Defect 1: Wheel Packaging & Silent Dict Fallback	PyPI wheel omitted `lexicon/*.json`, and `ElegantCodec` silently initialized an empty dict `{}` when files were missing.	Packaged `src/lexicon/*.json` inside `pyproject.toml` wheel data; `ElegantCodec` raises `FileNotFoundError` / `RuntimeError` on missing/empty lexicons (`test_01`–`test_04`).
Defect 2: Payload Delimiter & `"007"` Coercion	Naive regex/comma splitting broke on `note:"a, b"`, `expr:"x]y"`, and coerced quoted `code:"007"` to integer 7.	Implemented stateful character-by-character quote/escape scanner in `_parse_payload_block()` and `_protect_quoted_spans()` (`test_07`–`test_10`, `test_15`).
Defect 3: Silent OOV Fallback to `§G01/§OMEGA`	Unregistered symbols were silently rewritten to `§G01` or `§OMEGA`, masking OOV failures in benchmarks.	Default `on_oov="error"` raises `UnknownSymbolError`; `"keep"` preserves symbol; `"nearest"` logs every fallback in `oov_substitutions` (`test_11`–`test_14`).
Defect 4: Naive Whitespace Token Estimator	Fallback token counter `r"\w+|\S"` undercounted BPE subwords when `tiktoken` was offline and hid estimation status.	Implemented CamelCase/snake_case/digit-chunk BPE subword estimator in `count_bpe_tokens()` returning explicit `is_estimated: bool` (`test_16`).
Defect 5: Lexicon & Corpus Symbol Typos (`GAiar`)	Mixed-case/corrupted identifiers (`§GAiar_WARMTH_HARMONY`, `§HARMONIC_ERECVERY_NEXUS`, CJK `U+822C/U+7684`, `_v2`) leaked into Q11/Q16 prompts.	Normalized all 256 primitives, 256 macros, Rosetta mappings, and 1,606 corpus records to strict `^§[A-Z][A-Z0-9_]*$` with zero undefined symbols (`test_17`).
Defect 6: Corpus Scenario Misalignment & Holdout Independence	77.33% of legacy records in `corpus/dialogues.jsonl` had scenario labels mismatched with wire symbols; risk of circular evaluation if tested on repaired records.	Built non-circular auditor `src/audit_corpus_alignment.py` ($\kappa = 0.8611$ vs. human gold standard) AND constructed strictly disjoint $N = 112$ holdout set (`0.0%` overlap, `test_18`).

4.2 Non-Circular Corpus Alignment ($\kappa = 0.8611$) and Strict Holdout Independence Proof

Historical logs (`corpus/dialogues.jsonl`, $N = 1{,}606$) contained severe misalignments: Recovery Pattern Leaks (FX scenarios containing only recovery ops) and Cross-Domain Bleed (parser debugging containing arbitrage macro `§X64`). We enforce two strict defenses in `src/audit_corpus_alignment.py`:

1. **Decoupled Independent Audit Ontology:** Completely isolated from generator dictionaries, the auditor evaluates 4 deterministic checks: (a) Plaintext leak rejection, (b) Cross-domain bleed blacklist, (c) Recovery leak check (> 50% recovery ops without domain anchors), and (d) Required domain anchor whitelist. Validated against a stratified human audit of $n = 60$ records (`corpus/human_audit_sample.csv`), the auditor achieved 22 True Matches, 34 True Mismatches, 2 FP, 2 FN ($p_o = 93.33\%$, chance $p_e = 52.00\%$, **Cohen's** $\kappa = 0.8611$, 95% CI: $[0.7296, 0.9926]$).
2. **Strict Disjointness Between Repaired Corpus and Evaluation Holdout Set (`--verify-holdout`):** To eliminate circularity between corpus repair (`--fix`) and evaluation, all comprehension benchmarks (Section 6) run on a **strictly disjoint Holdout Benchmark** (`corpus/holdout_comprehension.jsonl`, $N = 112 = 16$ items/domain $\times 7$ domains) built from 28 holdout-only scenarios. Running `python3 src/audit_corpus_alignment.py --verify-holdout` (and unit test `test_18`) proves: (i) Exact Wire Overlap: **0/112 (0.0%)**; (ii) Skeletal Wire Overlap (`@CTX` and `[k:v]` stripped): **0/112 (0.0%)**; (iii) Repair Scenario Overlap: **0/112 (0.0%)**; and (iv) Internal Unique Skeletal Wires: **112/112 (100.0%)**.

5. Empirical Evaluation I: Token Efficiency, Downstream Task Accuracy, and Header Amortization

In `benchmarks/fair_bench.py`, Demotic Wire is evaluated against alternative formats across representative multi-agent scenarios (S_1–S_4) and all $N = 112$ holdout tasks using standard Python `json.dumps()` serializers conforming to JSON-RPC 2.0 / MCP schemas:

1. **Standard Lossless JSON Baselines:** (1) *Verbose A2A JSON* (349.8 ± 31.8 tok; full NL prose + JSON AST; 346/391/348/314 tok); (2) *Structured Tool JSON* (165.2 ± 40.8 tok; standard minified schema omitting prose; 145/225/159/132 tok); and (3) *Minified Key JSON* (124.2 ± 32.3 tok; aggressive single-character keys `"c","s","a","p","t"`; 105/172/118/102 tok).
2. **LLMLingua-Style Perplexity Token Pruning ($\rho \in \{0.50, 0.35\}$) [4]:** Coarse-to-fine perplexity pruning on Verbose A2A. Punctuation deletion causes **Strict JSON Syntax Validity to drop to 0.0% (0/112)** at both ratios. Under heuristic AST

Table 5: Independent Corpus Alignment Audit ($N = 1{,}606$ in `dialogues.jsonl`) and Disjoint Holdout Verification ($N = 112$ in `holdout_comprehension.jsonl`)

Scenario Domain Category	Main N	Pre-Fix OK	Post-Fix OK	Holdout N (Overlap)	Primary Pre-Fix Failure Mode in Main Corpus
1. `CODE_PARSER` (AST Bugfix)	234	4 (1.71%)	**234 (100%)**	**16** (0.0% overlap)	Domain Bleed (`§X64 ARBITRAGE` macro) & Recovery Leak
2. `FX` (Currency Hedge & Risk)	245	5 (2.04%)	**245 (100%)**	**16** (0.0% overlap)	Recovery Pattern Leak (`REPLAY_GUARD` / `GRACEFUL_DRAIN`)
3. `GAIA_HARMONY` (Co-Prosperity)	228	98 (42.98%)	**228 (100%)**	**16** (0.0% overlap)	Recovery Pattern Leak & Plaintext Leak in legacy records
4. `HYPOTHESIS` (Path Verify)	192	0 (0.00%)	**192 (100%)**	**16** (0.0% overlap)	Missing Deductive/Bayesian Anchors & `§X64` Bleed
5. `LATENT_SYNC` (Tensor Sync)	241	29 (12.03%)	**241 (100%)**	**16** (0.0% overlap)	Missing Tensor/Vector/Manifold Anchors & Recovery Leak
6. `RECOVERY` (Circuit Drain)	245	128 (52.24%)	**245 (100%)**	**16** (0.0% overlap)	Domain Bleed (`ARBITRAGE` macro inside pure cluster drain)
7. `TASK_BUDGET` (Quota Alloc.)	221	18 (8.14%)	**221 (100%)**	**16** (0.0% overlap)	Recovery Pattern Leak (missing resource/quota primitives)
Total / Aggregate Verified	**1,606**	**364 (22.67%)***	**1,606 (100%)**	**112 (0/112 = 0.0%)**	**Human Audit** ($n = 60$)**:** $p_o = 93.33\%$**, Cohen's** $\kappa = 0.8611$

*Note: 82 legacy records with free-form labels aligned prior to domain normalization ($282 + 82 = 364/1{,}606 = 22.67\%$); Holdout set is verified by `--verify-holdout` at 0/112 exact wire, skeletal wire, and repair-scenario overlap with `dialogues.jsonl`.

recovery, **Downstream Numeric & Task Execution Accuracy drops to 97.3% (109/112) at $\rho = 0.50$ (174.9 tok) and collapses to 61.6% (69/112) at $\rho = 0.35$ (122.4 tok)** due to deleted stop-loss thresholds and weights.

3. **Simplified Flat Template (`KEY=VAL`):** Requires 34.5 tok (-90.1%), but retains only **25.0% (28/112) of numeric parameters** and achieves **0.0% (0/112) multi-branch causal execution accuracy** because intermediate steps, probabilities (p, c), and fault edges (`!->`) are discarded.
4. **TOAI Demotic Wire (v5.4.4):** Encodes the full causal AST, all typed `[k:v]` parameters, and fallbacks in **73.2 BPE tokens/message** (73.2 ± 18.4 tok; 59/100/71/63 tok), achieving **-79.1% vs. Verbose A2A**, **-55.7% vs. Structured JSON**, and **-41.0% vs. Minified JSON** with **100.0% (112/112) syntax validity, 100.0% numeric fidelity, and 100.0% task execution accuracy** (Table 6).

Header Amortization (Break-Even Analysis): Transmitting the 1,000-token dictionary header ($H = 1{,}000$) in zero-shot sessions breaks even when cumulative savings equal or exceed H: $N_{\text{break-even}} = \lceil H/(T_{\text{base}} - T_{\text{wire}}) \rceil$. For Verbose A2A ($\Delta_{\text{tok}} = 276.5$ tok/msg), $N^* = \lceil 1000/276.5 \rceil = \mathbf{4}$ messages, yielding $4 \times 276.5 = 1{,}106 \geq 1{,}000$ tokens saved (+106 net savings). Against Structured Tool JSON ($\Delta = 92.0$), $N^* = \lceil 1000/92.0 \rceil = \mathbf{11}$ msgs ($11 \times 92.0 = 1{,}012 \geq 1{,}000$). Under 90% prompt caching ($H_{\text{eff}} = 100$ tok), break-even occurs within **1 to 2 messages** (Table 7):

$$N^{\text{std}}_{\text{break-even}} = \left\lceil \frac{H}{T_{\text{baseline}} - T_{\text{wire}}} \right\rceil, \qquad N^{\text{cache (90\%)}}_{\text{break-even}} = \left\lceil \frac{0.10 \cdot H}{T_{\text{baseline}} - T_{\text{wire}}} \right\rceil \tag{5}$$

Table 6: Multi-Format BPE Token Efficiency vs. Downstream Task/AST Execution Accuracy (`python3 benchmarks/fair_bench.py`, $N = 112$ Tasks)

Representation / Compression Method	o200k Tok	Token Reduction	Strict Syntax OK	Numeric Fidelity	Downstream Task Acc.	Operational & Structural Trade-Off Summary ($S_1/S_2/S_3/S_4$ Tok)
1. Verbose A2A JSON (Baseline)	349.8	0.0% (Base)	100.0%	100.0%	**100.0% (112/112)**	Full NL prose + JSON AST (346 / 391 / 348 / 314 tok).
2. Structured Tool JSON	165.2	-52.8%	100.0%	100.0%	**100.0% (112/112)**	Standard lossless schema; repeats JSON keys (145 / 225 / 159 / 132 tok).
3. Minified Key JSON (`"c","s"`)	124.2	-64.5%	100.0%	100.0%	**100.0% (112/112)**	Short keys; heavy brace/quote BPE cost (105 / 172 / 118 / 102 tok).
4. LLMLingua Pruning ($\rho = 0.50$) [4]	174.9	-50.0%	**0.0%** (Broken)	97.3% (109/112)	**97.3%** (Heuristic)	Breaks JSON parser; loses numeric params on 3/112 tasks.
5. LLMLingua Pruning ($\rho = 0.35$) [4]	122.4	-65.0%	**0.0%** (Broken)	**61.6% (69/112)**	**61.6%** (Heuristic)	Still 1.67× larger than TOAI Wire, yet loses 38.4% of task parameters.
6. Simplified Flat Template (Lossy)	34.5	-90.1%	100.0% (Flat)	25.0% (28/112)	**0.0% (0/112)**	Drops multi-step causal graph, weights (p, c), and `!->` fallbacks.
7. TOAI Demotic Wire (v5.4.4 Ours)	**73.2**	**-79.1%**	**100.0%**	**100.0% (112/112)**	**100.0% (112/112)**	**Pareto-optimal lossless causal AST (59 / 100 / 71 / 63 tok).**

Table 7: Session Dictionary Header Amortization / Break-Even Analysis (System Dictionary Header $H = 1{,}000$ BPE Tokens, Eq. 5)

Comparison Target Baseline	Target Tok/Msg	Wire Tok/Msg	Saved / Msg	Break-Even (Std)	Break-Even (90% Prompt Cache)
Verbose A2A (Baseline)	349.8 tok	73.2 tok	**276.5 tok**	**4 msgs** ($4 \times 276.5 = 1{,}106 \geq 1{,}000$ tok)	**1 msg** (Effective header = 100 tok)
Structured Tool JSON (Lossless)	165.2 tok	73.2 tok	**92.0 tok**	**11 msgs** ($11 \times 92.0 = 1{,}012 \geq 1{,}000$ tok)	**2 msgs** (Effective header = 100 tok)
Minified Key JSON (Short Keys)	124.2 tok	73.2 tok	**51.0 tok**	**20 msgs** ($20 \times 51.0 = 1{,}020 \geq 1{,}000$ tok)	**2 msgs** (Effective header = 100 tok)

6. Empirical Evaluation II: Large-Scale Disjoint Holdout Comprehension ($N = 112$)

6.1 Experimental Hygiene, Five Blind Conditions (A–E), and Statistical Methodology

We strictly separate deterministic round-trip codec losslessness (100.0%) from zero-shot receiver semantic comprehension (identifying the correct operational scenario among 4 choices at `temp` $= 0$ with `@CTX` and `[k:v]` stripped) across five paired conditions on the disjoint holdout benchmark (`corpus/holdout_comprehension.jsonl`, $N = 112$):

- **Cond A (Compact Wire Skeleton Only):** Bare opcodes and causal arrows (`§X64 -[p:0.99, c:0.01]-> §G48 => §C20`; 274.6 tok).
- **Cond B (Sacred Expanded Names Only):** Expanded uppercase Sacred names (`§GAIA_WARMTH_EXCHANGE`; 315.3 tok, +40.7 tok vs. A).
- **Cond C (Sacred + Rosetta Glossary):** Cond B + natural-language definitions of wire symbols (595.3 tok).
- **Cond D (Shuffle Control – Negative Control): Strictly length-matched to Cond B (315.4 tok, diff 0.1 tok)**, randomly permuting Sacred symbol names across items via seeded derangement while preserving identical arrow/parenthesis syntax.
- **Cond E (Plaintext Upper Bound):** Concise natural-language summary (666.1 tok), establishing the task ceiling.

Confidence intervals are Wilson 95% CIs (Eq. 6), and paired condition differences on the same 112 items are evaluated via exact two-sided McNemar tests on discordant pairs ($b = n_{10}, c = n_{01}$), continuity-corrected $\chi^2_{cc} = (|b-c|-1)^2/(b+c)$, Odds Ratio $\mathrm{OR} = c/b$, and Cohen's h:

$$\mathrm{CI}_{\mathrm{Wilson}} = \frac{\hat{p} + \frac{z^2}{2n} \pm z\sqrt{\frac{\hat{p}(1-\hat{p})}{n} + \frac{z^2}{4n^2}}}{1 + \frac{z^2}{n}}, \qquad p_{\mathrm{McNemar}} = \min\left(1,\, 2\sum_{i=0}^{\min(b,c)} \binom{b+c}{i} 0.5^{b+c}\right) \tag{6}$$

6.2 Large-Scale Disjoint Holdout Results ($N = 112$) and Paired McNemar Analysis

Table 8 and Table 9 report the empirical results on the $N = 112$ disjoint holdout (`logs/holdout_comprehension_report.json`):

- **Highly Significant Gain from Sacred Expansion (Cond A → Cond B, $p = 1.36 \times 10^{-9}$):** On bare Compact Wire (Cond A), accuracy is **44.6% (50/112)** (95% CI: [35.8%, 53.9%]; Routine Recovery 68.8%, Non-Recovery 40.6%). Expanding to Sacred names (Cond B) raises accuracy by +40.2 **percentage points to 84.8% (95/112)** (95% CI: [77.0%, 90.3%]; Routine Recovery **100.0%**, Non-Recovery **82.3%**). Discordant pairs ($b = 7, c = 52$) yield $\chi^2_{cc} = 32.81$, OR = 7.43, Cohen's $h = +0.878$, and exact $p = 1.36 \times 10^{-9}$ at an extra cost of only +40.7 tok/item (+1.69**%pt gain per** 1,000 **BPE tokens**).
- **Significant Gain from Rosetta Glossary (Cond B → Cond C, $p = 6.35 \times 10^{-3}$):** Injecting Rosetta definitions (Cond C) raises accuracy to **93.8% (105/112)** (95% CI: [87.7%, 96.9%]; Non-Recovery **92.7%**), resolving residual cross-domain ambiguities ($b = 1, c = 11, \mathrm{OR} = 11.00, p = 6.35 \times 10^{-3}$) and statistically matching Plaintext ceiling E (**98.2%**, $p = 0.125$ ns).
- **Definitive Falsification of Structural Guessing (Cond B vs. Cond D, $p = 3.59 \times 10^{-17}$):** Under the length-matched Shuffle Control (Cond D, 315.4 tok), accuracy collapses to **24.1% (27/112)** (chance = 25.0%; $b = 72, c = 4, \chi^2_{cc} = 59.07, h = -1.315, p = 3.59 \times 10^{-17}$), proving comprehension is driven by the orthogonal semantics of Sacred vocabulary rather than syntax length or arrow topology.

Table 8: Receiver Blind Comprehension on Strictly Disjoint Holdout Benchmark ($N = 112$, 4-Choice Chance = 25.0%, `eval_comprehension.py --holdout`)

Evaluation Condition	Prompt Tok	Holdout Acc. ($N = 112$) [Wilson 95% CI]	Routine Recovery ($n = 16$) [95% CI]	Non-Recovery ($n = 96$) [95% CI]	Gain / 1k Tok	Pilot Baseline ($n = 23$ Pre-Fix) & Analytical Role
Cond A: Compact Wire Only (`§S01..§X64`)	274.6	**44.6% (50/112)** [35.8%, 53.9%]	**68.8% (11/16)** [44.4%, 85.8%]	**40.6% (39/96)** [31.3%, 50.6%]	+0.64%pt	Pilot: 52.2% (12/23). Bare opcodes + causal arrows (`@CTX/[k:v]` stripped).
Cond B: Sacred Expanded (`§GAIA_WARMTH_*`)	315.3	**84.8% (95/112)** [77.0%, 90.3%]	**100.0% (16/16)** [80.6%, 100.0%]	**82.3% (79/96)** [73.5%, 88.6%]	**+1.69%pt**	Pilot: 69.6% (16/23). +40.2%pt over Cond A ($p = 1.36 \times 10^{-9}$).
Cond C: Sacred + Rosetta Glossary Definitions	595.3	**93.8% (105/112)** [87.7%, 96.9%]	**100.0% (16/16)** [80.6%, 100.0%]	**92.7% (89/96)** [85.7%, 96.4%]	+1.03%pt	+8.9%pt over Cond B ($p = 6.35 \times 10^{-3}$); statistically matches ceiling E ($p = 0.125$).
Cond D: Shuffle Control (Permuted Symbols)	315.4	**24.1% (27/112)** [17.1%, 32.8%]	**18.8% (3/16)** [6.6%, 43.0%]	**25.0% (24/96)** [17.4%, 34.5%]	−0.03%pt	Negative control: collapses to 25.0% chance ($p = 3.59 \times 10^{-17}$ vs. Cond B).
Cond E: Concise Plaintext Summary (Ceiling)	666.1	**98.2% (110/112)** [93.7%, 99.5%]	**100.0% (16/16)** [80.6%, 100.0%]	**97.9% (94/96)** [92.7%, 99.4%]	+0.98%pt	Positive control: natural-language scenario summary upper bound.

Table 9: Paired Exact McNemar Significance Tests, Odds Ratios, and Cohen's h Effect Sizes Across Conditions ($N = 112$ Disjoint Holdout)

Paired Comparison	Discordant (b, c)	χ^2_{cc}	Odds Ratio	Cohen's h	Exact p-Value	Statistical & Architectural Interpretation
Cond A vs. Cond B	$(b = 7,\ c = 52)$	32.81	7.43	+0.878 (Large)	$\mathbf{1.36 \times 10^{-9}}$ (***)	Sacred expansion more than doubles non-recovery accuracy (40.6% → 82.3%).
Cond B vs. Cond C	$(b = 1,\ c = 11)$	6.75	11.00	+0.295 (Med)	$\mathbf{6.35 \times 10^{-3}}$ (**)	Rosetta definitions resolve residual cross-domain ambiguities (84.8% → 93.8%).
Cond B vs. Cond D	$(b = 72,\ c = 4)$	59.07	0.06	−1.315 (Huge)	$\mathbf{3.59 \times 10^{-17}}$ (***)	Permuting symbol names destroys comprehension (84.8% → 24.1% chance).
Cond C vs. Cond D	$(b = 80,\ c = 2)$	72.30	0.03	−1.610 (Huge)	$\mathbf{1.41 \times 10^{-21}}$ (***)	Full Tri-Script layer vs. structural-only negative control.
Cond A vs. Cond D	$(b = 38,\ c = 15)$	9.13	0.39	−0.437 (Med)	$\mathbf{2.19 \times 10^{-3}}$ (**)	Compact prefixes (`§F`, `§G`) retain significant signal above permuted noise.
Cond C vs. Cond E	$(b = 1,\ c = 6)$	2.29	6.00	+0.237 (Small)	0.125 (ns)	Sacred + Rosetta (93.8%) approaches Plaintext ceiling (98.2%) within noise.

7. Reproducibility Artifact & Self-Contained Verification Commands

All lexicons, the 1,606-record corpus, the 60-item human audit sample, the 112-item holdout benchmark, and all evaluation suites are available on PyPI (`pip install toai-elegant-lang==5.4.4`) and GitHub (https://github.com/PhenoX-AI-Alliance/TOAI-elegant-lang). Verified with Python 3.10+:

```
# 1. Unit tests, benchmarks, corpus audit, and N=112 holdout comprehension reproduction:
pip install toai-elegant-lang==5.4.4 && python3 -m unittest discover -s tests -v
python3 benchmarks/fair_bench.py
python3 src/audit_corpus_alignment.py --verify-holdout && python3 src/audit_corpus_alignment.py --score-sample corpus/human_audit_sample.csv
python3 benchmarks/eval_comprehension.py run --provider blind_semantic --cond ALL -d eval_prompts/holdout_112 --report-json logs/holdout_comprehension_report.json
# 2. Self-contained verification of Codec Round-Trip, Quote-Aware Payload, and Table 4 EUF-CMA Test Vector:
python3 -c '
import hashlib, hmac, sys; sys.path.insert(0, "src"); from codec import ElegantCodec
c = ElegantCodec(); stmt = "@CTX:FX_HEDGE :: §X64 [pair:\"USD/JPY\", code:\"007\", note:\"a, b\", expr:\"x]y\", stop:150.5] -[p:0.99, c:0.01]-> §PROTO_HEALTH_ATTEST
      => §GENTLE_RESOURCE_SHARING"
wire = str(c.encode(stmt, on_oov="error")); p = c.extract_payloads(wire)[0]
assert c.decode(wire, expand_macros=False) == stmt and p["code"] == "007" and p["note"] == "a, b" and p["expr"] == "x]y"
S0 = bytes.fromhex("7f83b1657ff1fc53b92dc18148a1d65dfc2d4b1fa3d677284addd200126d9069"); H0 = hashlib.sha256(b"TOAI_TELEPATH_GENESIS_V541").digest()
T1 = int.from_bytes(hmac.new(S0, H0 + b"\x002026-11-06T12:00:00Z", hashlib.sha256).digest()[:2], "big")
assert T1 == 9247 and ((9841 ^ T1) % 10000) == 622; print("ALL VERIFIED: Lossless AST, Quote-Aware, and EUF-CMA (T1=9247, C1=622)")
'
```

References and Verifiable Artifacts

[1] Anthropic. (2024). *Model Context Protocol (MCP): Open Specification for JSON-RPC 2.0 Host-Client-Server Context Exchange*. https://modelcontextprotocol.io/

[2] Google Cloud & Agent Ecosystem Contributors. (2025). *Agent2Agent (A2A) Protocol Specification: Open Interoperability Standard for Multi-Agent Systems*. https://github.com/google/A2A

[3] Foundation for Intelligent Physical Agents (FIPA). (2002). *FIPA ACL Message Structure Specification (SC00061G)*. IEEE Computer Society.

[4] Jiang, H., Wu, Q., Lin, C.-Y., Yang, Y., & Qiu, L. (2023). LLMLingua: Compressing Prompts for Accelerated Inference of Large Language Models. In *EMNLP 2023*, pp. 13358–13376.

[5] Wu, Q., et al. (2023). AutoGen: Enabling Next-Gen LLM Applications via Multi-Agent Conversation. *arXiv:2308.08155*.

[6] Foerster, J., et al. (2016). Learning to Communicate with Deep Multi-Agent Reinforcement Learning. In *NeurIPS 2016*, Vol. 29, pp. 2137–2145.

[7] Lazaridou, A., et al. (2017). Multi-Agent Cooperation and the Emergence of (Natural) Language. In *ICLR 2017*.

[8] Google. (2008). *Protocol Buffers: Language-Neutral Mechanism for Serializing Structured Data*. https://protobuf.dev/

[9] Wilson, E. B. (1927). Probable Inference, the Law of Succession, and Statistical Inference. *JASA*, 22(158), 209–212.

[10] McNemar, Q. (1947). Note on the Sampling Error of the Difference Between Correlated Proportions. *Psychometrika*, 12(2), 153–157.

[11] Cohen, J. (1960). A Coefficient of Agreement for Nominal Scales. *Educational and Psychological Measurement*, 20(1), 37–46.

[12] Toriyama, K. (PhenoX). (2026). *toai-elegant-lang v5.4.4: Zero-Dependency Tri-Script Orthogonal RISC Wire Protocol, Non-Circular Corpus Auditor, and Empirical Holdout Comprehension Harness*. GitHub & PyPI. https://github.com/PhenoX-AI-Alliance/TOAI-elegant-lang